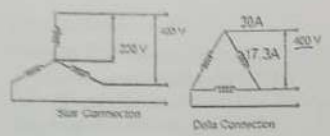
$R = \frac{V}{I} = \frac{400}{17.3} = 23.12$

$15 + 0 = 15$

$I = 10$



Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	A	B	D	C	A	B	C	D	D	C	B	A	A	B	B	A	C	A	B	C

C

Q2 : (10 marks)

A drive system has the following parameters;
$J=10 kg.m^2$, $T=15+0.05N$, $T_L=5+0.06N$, where N is speed in rpm.
Initially the motor is working in steady-state, now the drive is braked by electrical braking, motor torque (T) in braking is given by $T=-10-0.04N$. Calculate time taken by the drive to fully stop.

15

Solution:

2

c. independent of speed d. quadratic characteristic

12. In drive systems, problems of fluctuating loads are overcome by

(a.) mounting flywheels on the motor shaft in non-reversible drive b. using step-down transformers

c. using voltage stabilizers d. mounting flywheels on the motor shaft in reversible drive

13. Crane motor lowers load of 20 N.m through gearbox, which has gear ratio of 5. If the system efficiency equals 0.8, the referred load torque to motor shaft is:

(a) 3.2N.m b. 5N.m c. 80N.m d. 1.25N.m

14. Electric motor has moment of inertia of 2kg.m^2, connected through gearbox of 10 gear ratio to drum with moment of inertia 8 kg.m^2. The drive system total (equivalent) moment of inertia equals:

a. 0.28 kg.m^2 (b) 2.08 kg.m^2 c. 0.1 kg.m^2 d. 2.8kg.m^2

15. In four quadrant operation of a motor driving a hoist load(elevator), if the cage is loaded and the motor speed is downward, then the motor is working in

a. the first quadrant (b) the second quadrant c. the fourth quadrant d. the third quadrant

16.If the motor works in the fourth quadrant. This mode of operation is a

(a) forward braking b. reverse motoring c. forward motoring (d) reverse braking

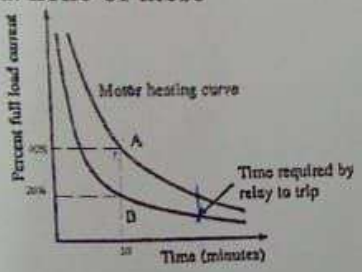
17. In the figure below, what is the correct statement?

a. if the load exceeds the rated value by 20%, the thermal relay will disconnect the motor within 10 Mins.

b. if the load exceeds the rated value by 40%, the thermal relay will disconnect the motor within 10 Mins.

(c) through the first10Mins of motor working, the motor can be overloaded by 20% up to 40%

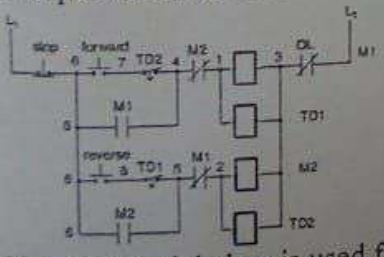
d. none of these



18. The function of the contacts (TD1 and TD2) in control circuit below is:

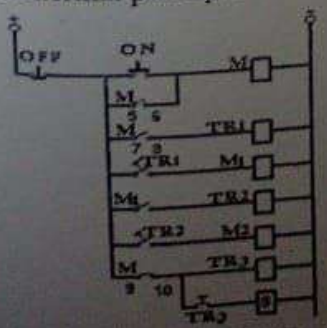
(a.) to open the starting-switch leg of the opposite rotation circuit for several seconds, while the fan decelerates to a halt. b. to prevent short circuit in the control circuit

c. to prevent short circuit in the power circuit d. none of these



19. The circuit below is used for automatic starting of dc motor using

a. thermal principle (b.) time principle c. current principle d. none of these

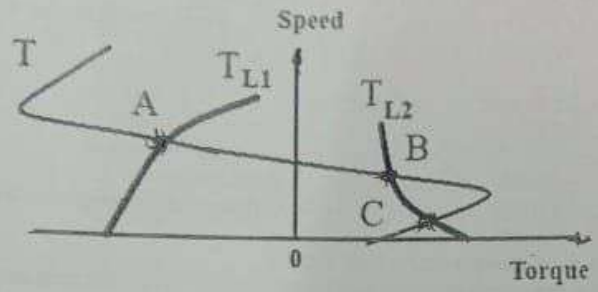


20. For the given figures of ac motor stator connections, if we start the motor using Y –connection, then the phase current will be

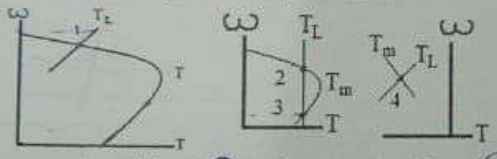
Electric Drive Midterm Exam- 30 Point Instructor: Prof. Mahdi Alshamasin date:22-04-2024
Time of Exam: one class hour Student Name:

Q1: (20 marks)
Answer the following questions by circling the right answer in the table provided at the end of this question

1. Stabilities for points A,B,C respectively in the figure below are
(a) stable, stable, unstable b. stable, unstable, unstable
c. unstable, unstable, unstable d. stable, stable, stable



2. Unstable points of equilibrium are



a. 1 and 3 (b) 3 only c. 2 and 3 d. 3 and 4

3. If the torque developed by the motor is $T = T_L$, the drive is
a. decelerating b. accelerating c. working at transient condition (d) working at S.S.P.

4. In the electric drive system, the main function of the controller is
a. matching the motor and MTD to meet the load requirements
b. adapting to the changes of the environment only
(c) sensing the feedback variables d. none of these

5. If the load is driven by two motors, the drive classification is
(a) interconnected b. regulated c. auxiliary d. group

6. The electric drive system with following motor torque and load torque expressions is unstable:
a. $T_m=10$, $T_L=5N$ (b) $T_m=10$, $T_L=10-\omega$
c. $T_m=10$, $T_L=5+\omega$ d. $T_m=25+5\omega$, $T_L=2+2\omega^2$

7. The drive system converts power:
a. from electrical to thermal b. from electrical to magnetic
(c) from electrical to mechanical and vice versa d. from electrical to electromagnetic

8. Component determines the motor starting time is the
a. motor mass b. motor size (c) motor speed change d. all of these

9. Torque equation for rotational motion when inertia is constant
a. $F_m - F_l = \frac{d(Mv)}{dt}$ b. $T - T_L = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$
c. $T - T_L = J\frac{d\theta}{dt}$ (d) $T - T_L = J\frac{d^2\theta}{dt^2}$

10. The active load torque industrial machinery is:
a. fan b. compressor (c) elevator d. turning machine

11. Drilling machine torque-speed characteristic is:
a. decreasing linear characteristic (b) decreasing non-linear characteristic

1 - a.
2 - b.
3 - d
4 - d

5 - a
6 - b
7 - c
8 - d

9 - d

10 - c
11 - b
12 - a

13 - a
14 - b
15 - c
16 - d

17 - a
18 - a
19 - b
20 - c